



DEPARTMENT OF AGRICULTURE AND BIOSYSTEMS ENGINEERING

BALAPOLOSI BOTLHE MORWAMANG

IRRIGATION WATER SUPPLY AND CONVEYANCE



INTRODUCTION

Surface water monitoring is an important component of water resource management as it provides the relations of water supply, flood mitigation, irrigation and environmental protection. In semi-arid regions such as Botswana, where rainfall is highly variable in both space and time and perennial river flow is limited, accurate measurement of streamflow is important. Long term hydrological records are essential for effective water resources planning, infrastructure design and sustainable catchment management.

This report focuses on the Kolobeng Weir which is part of the Notwane River catchment system. The Kolobeng River Weir gauging station is a hydrometric structure located along the Kolobeng River near the village of Kumakwane, approximately 25 km west of Gaborone. The purpose of the visit was to know about streamflow measurement techniques, hydrometric equipment and the operational principles underlying a weir-based gauging station.



METHODOLOGY

SITE DESCRIPTION

WEIR STRUCTURE

The hydraulic structure observed at the site is a rectangular sharp-crested weir, complemented by a concrete-lined intake canal and a narrow walkway bridge to facilitate measurement and maintenance access. The weir functions as the primary flow control and measurement structure, directing all stream discharge through a defined cross-section.

The intake canal channels water from the main river to the stilling well chamber, where a gauge plate is permanently installed. A sluice gate arrangement is also present, enabling flow regulation when necessary.



The photo above shows a rectangular weir.

There is also a portable water quality meter and also known as a multi-parameter probe. It is used for measuring pH, Electrical Conductivity (EC) and Dissolved Oxygen (DO).



The photos above show a multi-parameter meter.

PHYSICAL SETTING

The gauging station is established within a narrow, rocky gorge of the Kolobeng River, set against a backdrop of red sandstone geology and densely vegetated hillsides. The channel at this point is well-constrained between rock outcrops, providing hydraulic stability that is conducive to accurate flow measurement. The site lies at a gauge datum elevation of 6.588 m above an established reference level.

The surrounding environment is characterized by indigenous riparian vegetation, scattered boulders, and exposed bedrock, all of which contribute to the natural aesthetics while also presenting logistical challenges for field measurements. Access to the site is managed through a designated footpath leading to the measurement structures.



FLOW MEASUREMENT PRINCIPLES AND EQUIPMENT

Gauge Plate and Water Level Reading

A graduated gauge plate is installed at a fixed reference position within the stilling well adjacent to the weir. All water level readings are referenced to this gauge plate datum. During the field visit, it was established that meaningful flow readings begin at a gauge height of 1.6 m and above. Readings below 1.5 m are considered below the minimum detectable threshold and cannot be reliably related to discharge.

The gauge plate is consistently regarded as more accurate than staff gauge readings due to its controlled hydraulic environment within the stilling well.

The gauge plate reading is subsequently related to discharge using a rating curve. A site specific relationship between water stage (H) and discharge (Q) that has been established through repeated current meter gauging over multiple seasons. This conversion is stored within the HISTRA system, enabling automated transformation of stage data into discharge estimates.



The photo above shows how the gauge plate gets reading from the weir.

Sharp-Crested Weir Discharge Equation

The rectangular sharp-crested weir operates on the principle of critical flow at the crest. The standard discharge equation applied is:

$$Q = C \cdot L \cdot H^{3/2}$$

Where:

- Q = Volumetric discharge (m^3/s)
- C_d = Discharge coefficient (dimensionless, typically ~ 0.611 for a sharp-crested weir)
- L = Width of the weir notch (m)
- H = Head of water above the weir crest (m)

Current Meter (Hyquest Solutions)

A Hyquest Solutions electromagnetic current meter (Australia, model OSS-EL 20-10) was demonstrated during the practical. The instrument consists of a streamlined stainless-steel body

housing an electromagnetic velocity sensor, mounted on a graduated wading rod. An electromagnetic sensor measures water velocity by detecting the voltage induced as flowing water passes through the sensor's magnetic field, providing precise point velocity measurements without moving mechanical parts.

A red-cabled data cable connects the sensor to a handheld digital readout unit, enabling real-time velocity readings. The device is lowered to specified depths within the river cross-section to capture vertical velocity profiles. Multiple depth velocity readings across the channel width are then integrated to produce a mean cross-sectional discharge estimate.



A stilling well is installed at the gauging station to dampen surface wave effects and provide a stable water surface for accurate level measurement. The stilling well is connected to the river via an intake pipe, and a pressure transducer or float-based sensor is positioned within it. The data logger case observed in the field housed a compact digital logger unit with red and grey signal cables, calibration accessories, and storage provisions, consistent with a portable hydrometric deployment kit. Such systems can be connected to telemetry to enable remote data transmission to the national HISTRA database.

The Kolobeng River Weir gauging station contributes streamflow data to the national HISTRA hydrological archive maintained by the Department of Water and Sanitation. Continuous stage and discharge records collected at stations such as this are fundamental to a range of water resources applications, including:

- Flood frequency analysis and risk mapping for downstream communities
- Water allocation planning and abstraction licensing
- Assessment of catchment yield to support the feasibility of storage infrastructure such as the proposed Kolobeng Dam
- Ecological flow determination for riparian habitat protection

- Calibration and validation of hydrological models used in climate change impact assessment

The location of the station in the Kweneng District places it within the catchment that has historically been identified as a potential supplementary water source for the Greater Gaborone area, particularly in relation to studies carried out alongside the Metsimotlhabe River scheme. Accurate and consistent streamflow records from this site are therefore of strategic national importance.

Maintenance and Operational Considerations

Effective operation of the gauging station requires routine maintenance and periodic calibration. Key maintenance tasks identified during the field visit include:

- Cleaning of the weir crest and upstream approach channel using cleaning rods to remove debris, algae, and sediment accumulation that can distort the effective flow width
- Inspection and servicing of the sluice gate mechanism to ensure free operation
- Calibration of the pressure transducer and data logger against manual staff gauge readings
- Vegetation clearance around the stilling well and intake pipe to prevent blockage
- Structural inspection of the concrete weir body, canal lining, and walkway for signs of cracking or settlement

CONCLUSION

The field visit to the Kolobeng River Weir gauging station provided a comprehensive practical exposure to the principles and methods of surface water monitoring in Botswana.

The rectangular sharp-crested weir, operating in conjunction with the gauge plate, stilling well, and electromagnetic current meter, constitutes a well-equipped hydrometric installation capable of delivering reliable discharge data across a range of flow condition

Key takeaways from the practical include the critical role of the gauge plate as the primary measurement reference (with valid readings above 1.6 m), the application of the weir discharge equation in translating stage to flow, and the importance of the HISTRA system in aggregating national streamflow data.

The station's contribution to understanding the Kolobeng catchment hydrology positions it as a valuable asset in the context of Botswana's ongoing water security planning.

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